Project 1: Launch
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Research Question:

How did the spatial distribution of angling pressure in Montana change from 2019 to 2020 during the onset of the COVID-19 pandemic in relation to the proximity of population centers and the popularity of fishing rivers?

Hypothesis:

During the onset of the COVID-19 pandemic in 2020, angling pressure from residents and non-residents became more dispersed across rivers, with angling pressure increasing on less-popular rivers compared to 2019. Angling pressure also increased disproportionately on rivers located closer to population centers compared to 2019.

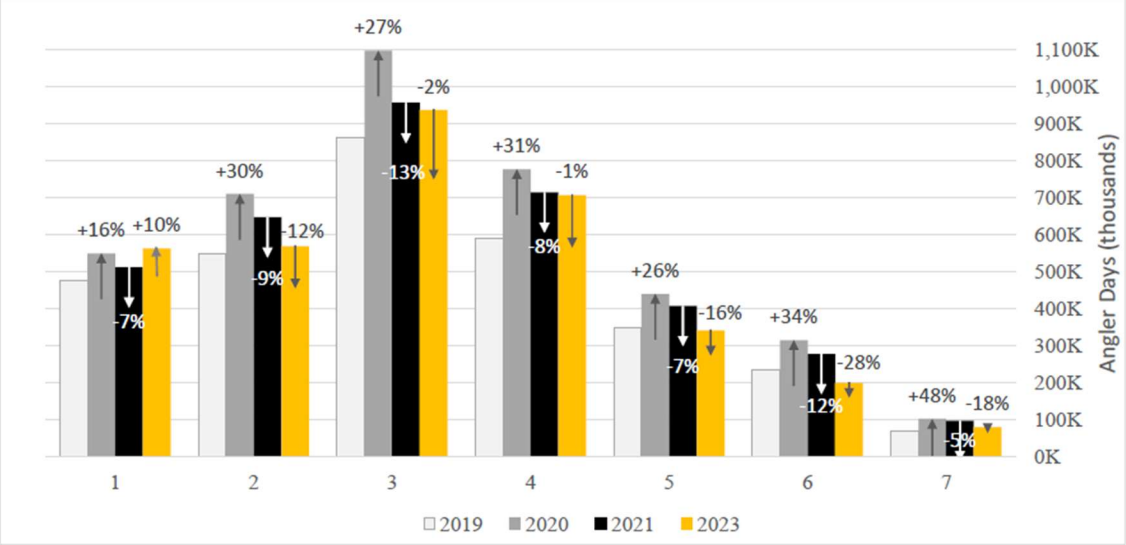


Figure 8. Percent change of annual angling pressure by region between 2019-2023 (residents and non-residents)

*We can see that angling pressure increased from 2019 to 2020, where did these increases happen?

Data Sources:

Vector

- **Montana State Library: National Hydrography Dataset (NHD)** (link)
    - Polyline and polygon data showing lakes, rivers, and streams in Montana
- **Fish, Wildlife & Parks Fisheries Regions** (link)
    - FWP administrative boundaries
- **2019 & 2020 Montana Statewide Angling Pressure Summary Report**

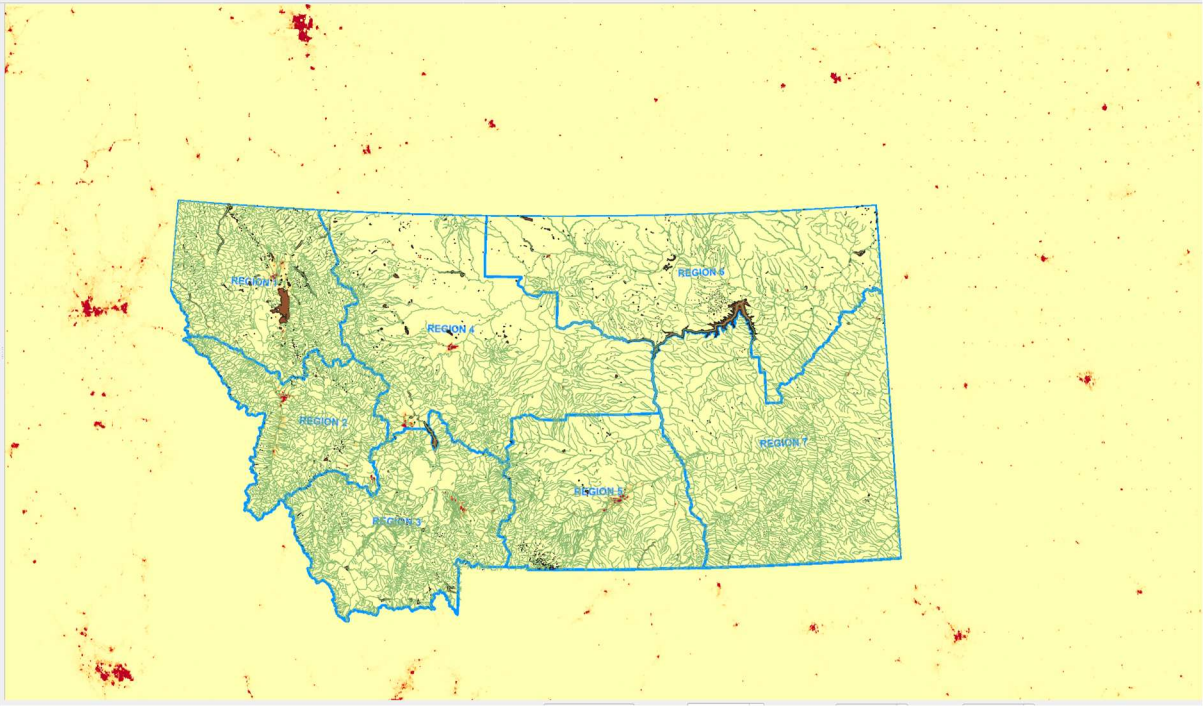
Table 6. Angling Pressure in angler days by Drainage by Lake or Stream for the 2020 survey license year

	--- Totals ---		--- Resident ---		--- Non-Resident ---	
	Pressure	Trips	Pressure	Trips	Pressure	Trips
Blackfoot River						
Lake	55,349	393	46,786	341	8,563	52
Stream	112,152	814	77,524	587	34,628	227
Total:	167,501	1,207	124,310	928	43,191	279

*Scrape and append data from reports to NHD dataset

Raster

- Gridded Population of the World v4.11, Population Density (NASA SEDAC) – population proximity variables (link)
    - Population density index



Measurement Methods:

- Total angling pressure values from the angling surveys will be joined with the NHD layer
- Use a color ramp to display which rivers experienced the greatest increase in angling pressure
- Rivers falling within 50 miles of population centers will be considered high-proximity

Potential Outputs:

- Bar graph showing the greatest increase in angling pressure
- Regional comparisons

Exploded maps to look more closely at populated areas in Montana